Lesson 6 Teacher's Key

GameMaker

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

obj_enemy's built-in Destroy event does the counting, reaching out to a dedicated obj_gamemanager instance rather than the spawner:

```
// obj_spawner's Create event (spawning only, unchanged from Lesson 5):
for (var i = 0; i < 5; i++) {
    instance_create_layer(80 + i * 96, 80, "Instances", obj_enemy);
}

// obj_gamemanager's Create event (one instance, placed once in the room):
enemies_remaining = 5;
has_won = false;

// obj_enemy's Destroy event:
obj_gamemanager.enemies_remaining -= 1;
if (obj_gamemanager.enemies_remaining <= 0) {
    obj_gamemanager.has_won = true;
}

// obj_gamemanager's Draw GUI event:
if (has_won) {
    draw_text(400, 50, "You Win!");
}
```

Level 2 - Complete Worked Solution

GameMaker's Destroy event fires exactly once per instance, no matter what caused the destruction, and obj_gamemanager's has_won only ever flips from false to true - there's no path where the win text could show early, late, or more than once.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on win

Import the provided sfx_win.wav as a new sound resource (keep its default name, sfx_win), and play it exactly once, right when has_won first flips true:

```
// obj_enemy's Destroy event:
obj_gamemanager.enemies_remaining -= 1;
```

```
if (obj_gamemanager.enemies_remaining <= 0 && !obj_gamemanager.has_won) {
    obj_gamemanager.has_won = true;
    audio_play_sound(sfx_win, 1, false);
}
```

Visual: styled win text

Center and enlarge the text using draw_set_halign() and draw_set_font():

```
// obj_gamemanager's Draw GUI event:
if (has_won) {
    draw_set_halign(fa_center);
    draw_text_transformed(400, 50, "You Win!", 2, 2, 0);
}
```

Structure: obj_gamemanager owns the counting

All win-tracking (enemies_remaining, has_won) lives as variables on obj_gamemanager - obj_spawner only ever spawns enemies, and obj_enemy's Destroy event only ever reaches out to modify obj_gamemanager's variables. None of the three draws anything except obj_gamemanager itself.

Complete Level 3 solution (all 2 goals combined)

The Destroy event and Draw GUI event with both ideas added:

```
// obj_enemy's Destroy event:
obj_gamemanager.enemies_remaining -= 1;
if (obj_gamemanager.enemies_remaining <= 0 && !obj_gamemanager.has_won) {
    obj_gamemanager.has_won = true;
    audio_play_sound(sfx_win, 1, false);
}

// obj_gamemanager's Draw GUI event:
if (has_won) {
    draw_set_halign(fa_center);
    draw_text_transformed(400, 50, "You Win!", 2, 2, 0);
}
```

Alternate Path Notes: GML Visual

obj_enemy's Destroy event works identically whether it holds blocks or code - the event itself already fires exactly once per instance either way, and reaches out to obj_gamemanager the same way in both paths.

The Level 3 goals map onto blocks the same way: an Audio category block gated by an "If Variable" checking has_won, and "Draw Text" blocks with transform options for the visual idea, both still living on obj_gamemanager.

- Keeping the same Structure principle (obj_gamemanager owns counting, obj_spawner only spawns, obj_enemy only reports) whether a student is in Code or Visual reinforces that it's a design idea, not a syntax rule.